

Tribhuvan University
Institute of Science and Technology

A Dissertation Proposal on

**“Comparative Benchmarking of CatBoost, TabPFN, and
TabFM for Explainable Multi-Tier Hair Fall Risk
Stratification”**

Supervised by
Asst. Prof. Jagadish Bhatta

Submitted by
Nirajan Shahi
Roll no. 49/079

Submitted to
Central Department of Computer Science and Information Technology
Tribhuvan University, Kirtipur
Kathmandu, Nepal

In partial fulfillment of the requirement for Master’s Degree in Computer Science and
Information Technology (M.Sc. CSIT)

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List of Abbreviations

ALT	Alanine Aminotransferase
ARIMAX	Autoregressive Integrated Moving Average with Exogenous Variables
AUC	Area Under the Curve
AutoML	Automated Machine Learning
CNN	Convolutional Neural Network
COVID-19	Coronavirus Disease 2019
DT	Decision Tree
EHR	Electronic Health Record
GB	Gradient Boosting
GPU	Graphics Processing Unit
KNN	k-Nearest Neighbours
LR	Logistic Regression
LSTM	Long Short-Term Memory
RF	Random Forest
ROC	Receiver Operating Characteristic
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SHAP	Shapley Additive Explanations
SVM	Support Vector Machine
TabFM	Tabular Foundation Model
TabPFN	Tabular Prior-Data Fitted Network
TFT	Temporal Fusion Transformer
XGBoost	Extreme Gradient Boosting

Chapter 1: Introduction

1.1 Introduction

Alopecia, the clinical term for pathological hair loss, affects individuals across a broad range of ages and demographic backgrounds worldwide. Some degree of hair shedding is a normal part of the follicular growth cycle, but a marked or sudden increase in shedding often points to an underlying disturbance in the body's systems, commonly hormonal imbalance, significant nutrient depletion, prolonged physical strain, or sustained psychological stress (Bai et al., 2026; Guo & Katta, 2017; Hussein et al., 2023). Beyond its visible effects, noticeable hair loss frequently carries a real emotional cost, contributing to lowered self-confidence, elevated anxiety, and a reduced sense of overall well-being. In this proposal, "hair loss" and "hair fall" are used with the same meaning, and "hair fall" is also the name of the target variable in the dataset.

Determining an individual's likelihood of experiencing hair loss is inherently difficult because no single biomarker reliably predicts it. Risk instead emerges from the combined, non-linear interaction of physiological traits, everyday habits, environmental factors, and pre-existing health conditions. Elements such as diet quality, sleep regularity, scalp care practices, thyroid activity, iron levels, and hereditary predisposition all interact at once, making it impractical for clinicians or individuals to gauge overall risk through observation alone. Chapter 2 describes these biological causes in more detail.

Healthcare systems today generate large volumes of patient data through Electronic Health Records (EHRs), capturing detailed clinical histories over time. A substantial share of this clinical data is inherently tabular in structure: laboratory results, vital signs, diagnostic codes, medication histories, and patient demographics are all naturally represented as structured rows and columns rather than as free text or medical imagery. Consequently, tabular data forms the backbone of a large fraction of real-world clinical prediction tasks, including disease risk stratification, treatment outcome prediction, and, as in this study, early screening for conditions such as hair loss. Machine learning offers a way to surface patterns within this dense, multi-source tabular data that would otherwise go unnoticed. Gradient boosting methods have long been the standard tool for prediction on structured, tabular data (Prokhorenkova et al., 2018). More recently, however, tabular foundation models, which are

transformer architectures pretrained to perform prediction directly from context without task-specific training, have begun to reshape this space. Prior Labs' TabPFN (Hollmann et al., 2025) and Google's TabFM (Kong et al., 2026) both represent this newer generation of zero-shot tabular models, offering predictions through a single forward pass rather than the conventional train-and-tune cycle.

A persistent limitation of both gradient boosting and transformer-based approaches, however, is their tendency to operate as opaque, uninterpretable systems. This research addresses that limitation by building a comparative machine learning framework for evaluating and explaining multi-tier hair fall risk, using structured clinical and lifestyle data. The study benchmarks three distinct modelling paradigms: CatBoost, a gradient-boosted decision tree architecture purpose-built for categorical data; TabPFN, an established prior-fitted transformer foundation model for tabular in-context learning; and TabFM, Google Research's more recently released zero-shot tabular foundation model. Post-hoc explainability is layered across all three models via Shapley Additive Explanations (SHAP; Lundberg & Lee, 2017), giving each prediction a transparent, feature-level justification grounded in the underlying medical and lifestyle indicators.

1.2 Problem Statement

Determining hair loss risk remains a genuinely difficult clinical problem, shaped by the interplay of hormonal shifts, metabolic irregularities, chronic stress, nutritional gaps, and hair care habits. Current practice leans heavily on subjective self-reporting or delayed clinical evaluation, both of which are vulnerable to observer bias, inconsistent standards, and limited reach in under-resourced settings.

Machine learning can, in principle, model these interacting risk factors systematically, but the literature reviewed in Chapter 2 leaves five problems open. First, studies that predict hair loss from survey and clinical tables have compared only classical algorithms such as Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost), on different datasets, and their reported accuracies range from about 50% to 100% (Khatun et al., 2022; Kumar et al., 2025; Siami & Azis, 2025), so it is not clear which type of model suits this problem. Second, CatBoost, which was designed for categorical data, appeared in only one hair loss study as a side comparison with a low score, apparently without tuning (Kumar et al., 2025). Third, tabular foundation models such as TabPFN and TabFM have been

evaluated only on general benchmarks (Hollmann et al., 2025; Kong et al., 2026), and none has been tested on hair loss data. Fourth, TabFM is very new and has been reproduced independently only once, with software and memory problems reported (Pandey, 2026), so its practical limits are unclear. Fifth, the hair loss studies concentrate on accuracy, and none compares how a boosted-tree model and foundation models explain their predictions.

These gaps matter because a risk tier is useful to a clinician or a patient only if it is reliable and can be explained. This study therefore builds and benchmarks an explainable machine learning framework for multi-tier hair fall risk classification. By evaluating CatBoost against two independently developed tabular foundation models, TabPFN and TabFM, alongside SHAP-based feature attribution, it examines whether zero-shot, prior-fitted transformer approaches can match a carefully tuned gradient-boosted tree on tabular health data, while keeping every predicted risk tier traceable to specific, interpretable indicators.

1.3 Objectives

This dissertation has two objectives:

1. To implement and benchmark CatBoost, TabPFN, and TabFM for predicting multi-tier hair fall risk from structured clinical and lifestyle data.
2. To provide transparent, feature-level decision-support insights for every model through SHAP-based explainability, so that each predicted risk tier can be traced to specific clinical and lifestyle indicators.

The rest of this proposal follows from these objectives. Chapter 2 explains the biology of hair loss and reviews earlier studies, and ends with the research gaps that this study fills. Chapter 3 describes the datasets, the preprocessing, the three models, the SHAP method, the tools, and the evaluation metrics that will be used to meet the two objectives. Chapter 4 states the outcomes that are expected and gives the working schedule.

Chapter 2: Background Study and Literature Review

This chapter builds the case for the study. Section 2.1 explains what causes hair loss and how it affects people, which is the reason a risk model needs both blood measurements and lifestyle information. Section 2.2 reviews the earlier studies that tried to predict hair loss and the machine learning methods this study uses, and Section 2.2.1 ends with the research gaps that the objectives in Chapter 1 respond to.

2.1 Background Study

Human hair grows in a repeating cycle of three phases: a growth phase (anagen), a short regression phase (catagen), and a resting phase (telogen) after which the hair is shed. Losing some hair every day is therefore normal. Hair loss becomes a medical problem when this cycle is disturbed, so that too many follicles enter the resting phase early or the follicles shrink and stop producing thick hair. Clinically, the most common forms are androgenetic alopecia (pattern hair loss driven by hormones and inheritance), telogen effluvium (a temporary but heavy shedding that follows a physical or emotional trigger), and alopecia areata (patchy loss caused by an autoimmune reaction).

The biological factors that disturb the hair cycle are well documented, and they usually act together rather than alone. The main ones are described below.

Hormones and heredity. Androgenetic alopecia depends on inherited sensitivity of the follicles to androgen hormones, which is why family history is one of the strongest risk indicators. Agaoglu et al. (2021) studied how common early-onset androgenetic alopecia is and how it relates to lifestyle and dietary habits.

Nutrition and iron status. Hair follicles are among the most active tissues in the body, so they are sensitive to shortages of iron, zinc, and some vitamins. Guo and Katta (2017) reviewed the evidence on nutrient deficiency, supplement use, and hair loss. Lin et al. (2023) focused on iron-deficiency-related alopecia in women, and Treister-Goltzman et al. (2022) combined the available studies in a systematic review and meta-analysis of iron deficiency and non-scarring alopecia in women. These studies are why blood indicators such as haemoglobin and iron are treated as important risk markers.

Thyroid function. Thyroid hormones help regulate the hair cycle, and both an underactive and an overactive thyroid are linked with hair disorders (Hussein et al., 2023). In Nepal, Marahatta et al. (2018) reported on the association between alopecia areata and thyroid dysfunction in patients from eastern Nepal.

Psychological stress. Stress is thought to push follicles out of the growth phase too early, which leads to increased shedding some weeks later (Bai et al., 2026). Stress can also make an existing condition worse, so it is both a cause and a consequence of hair loss.

Infection and physical illness. A sudden illness, fever, or major physical strain can trigger telogen effluvium. Cline et al. (2021) documented a surge in telogen effluvium in communities heavily affected by coronavirus disease 2019 (COVID-19), showing how a large health event can raise the number of people who present with hair loss.

How people are affected in real life. Hair loss is rarely only a cosmetic matter. It is linked with lower self-confidence, anxiety, and a poorer quality of life, and in many cases the person does not know which of the factors above is responsible. Survey evidence shows that young people are affected too. In a survey of 610 participants in Bangladesh, most respondents were aged 18 to 24, and stress, allergies, dandruff, and family history were the main causes they reported (Khatun et al., 2022). Worldwide, the burden of alopecia areata has been analysed over three decades using the Global Burden of Disease data, including differences in lifetime risk between women and men (Sun et al., 2025).

Current trend. Reports suggest that hair loss is being seen more often and at a younger age. Early-onset androgenetic alopecia has been linked to lifestyle and dietary habits (Agaoglu et al., 2021), and post-infection shedding rose sharply during the COVID-19 period (Cline et al., 2021). Research and public interest have followed the same direction: bibliometric work has mapped global research trends and emerging topics in hair loss treatment (Wang et al., 2026), and Google search patterns have been used to track worldwide interest in hair loss treatments (Todorova & Kluger, 2026).

From biology to machine learning. Because the causes above overlap and interact, no single test can predict hair loss risk. Diagnosis today relies mostly on clinical examination and the patient's own history, which is slow, subjective, and hard to reach in rural areas. This has led researchers to apply machine learning to hair loss. Earlier work mainly used scalp or hair images (Shakeel et al., 2021; Sayyad et al., 2022), and more recent work uses survey and

clinical tables with classical algorithms such as Random Forest and XGBoost (Khatun et al., 2022; Kumar et al., 2025; Sai et al., 2023). The newest direction is tabular foundation models such as TabPFN (Hollmann et al., 2025) and TabFM (Kong et al., 2026), which have not yet been tested on hair loss. Section 2.2 reviews this work in detail.

2.2 Literature Review

This section reviews the published research that is closest to the present study. For each study it states what data was used, which models were tested, what was found, and what was left open. The studies are grouped by topic, and Table 2.1 at the end of the section summarises them.

Hair loss prediction from survey and lifestyle data. Khatun et al. (2022) surveyed 610 people in Bangladesh and trained a Support Vector Machine (SVM), k-Nearest Neighbours (KNN), Logistic Regression, Random Forest, and XGBoost to diagnose hair fall disorder. XGBoost gave the best accuracy of 92.62%. The study shows that questionnaire data can carry a useful signal, but it tested only classical models and did not explain individual predictions.

Sai et al. (2023) compared SVM, KNN, decision tree, random forest, and logistic regression with an ensemble method for hair fall prediction. The ensemble was better than every single algorithm in accuracy, precision, and recall. The study did not include boosting models designed for categorical data or any explanation method.

Kumar et al. (2025) built a Random Forest hair loss predictor on a dataset of 2,000 records with 10 features (genetics, hormones, medical history, nutrition, stress) and added a web interface built with Django. Random Forest reached 100% accuracy, while XGBoost (67.5%), CatBoost (49.5%), and LightGBM (47.5%) scored much lower. A score of 100% on a real health problem is unusual and may point to an easy or synthetic-looking dataset, so it is better read as a result on that one dataset than as proof that Random Forest is best. The very low CatBoost score also suggests that the boosting models were not tuned, which is an open question this study tests properly.

Siarni and Azis (2025) compared Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and a voting ensemble on a balanced dataset of genetic, hormonal, lifestyle, and environmental indicators. Accuracy and F1-score stayed at about

50% at best, although age, stress, and nutritional deficiency were identified as important factors. The authors concluded that richer clinical inputs were needed.

Leema et al. (2025) collected questionnaire data from 750 university students and community members and proposed HairSentinel, which forecasts hair fall trends and flags unusual episodes using a Temporal Fusion Transformer (TFT). It reached 97.5% accuracy and was compared with a Long Short-Term Memory network (LSTM), Random Forest, and an Autoregressive Integrated Moving Average with Exogenous Variables model (ARIMAX). Their dataset is not public, which makes the result impossible to reproduce, and it rests on self-reported data rather than blood test values.

Hair loss detection from images and deep learning. Shakeel et al. (2021) proposed a framework that classifies healthy hair and alopecia areata from hair images using colour, texture, and shape features with SVM and KNN, with accuracies of 91.4% and 88.9%. Sayyad et al. (2022) used a VGG network with an SVM classifier and reported 98.31% accuracy on 200 healthy-hair images (Figaro1k) and 68 alopecia areata images (DermNet). Pandikumar et al. (2024) proposed a deep learning framework that combines convolutional neural networks (CNNs) on scalp images with LSTM networks on lifestyle sequences. These studies need images or specialised equipment, address alopecia areata or scalp condition rather than risk from routine health indicators, and use small image sets.

Gradient boosting and CatBoost. Prokhorenkova et al. (2018) introduced CatBoost, which handles categorical features with ordered target statistics and ordered boosting to avoid target leakage. This design makes CatBoost a strong choice for survey-style data with many categorical answers. In hair loss research, however, CatBoost appears only as a side comparison and was not tuned (Kumar et al., 2025).

Tabular foundation models. Hollmann et al. (2025) presented TabPFN, a transformer pretrained on millions of synthetic datasets that predicts on a new table in a single forward pass without task-specific training. It performed best on small datasets of up to about 10,000 rows. Kong et al. (2026) introduced TabFM, a 400-million-parameter model trained on synthetic tables, which ranked first among default tabular foundation models on the 51 datasets of the TabArena benchmark and outperformed tuned automated machine learning (AutoML) pipelines. An independent reproduction on three machines and 13 TabArena datasets confirmed competitive accuracy against XGBoost, Random Forest, and TabPFN, but

reported four upstream software defects and memory failures above about 10,000 rows on a 24 GB graphics processing unit (GPU) (Pandey, 2026). None of these studies used medical or hair loss data.

Explainability. Lundberg and Lee (2017) proposed SHAP, which assigns each feature a contribution to a prediction using Shapley values from game theory. Because SHAP works with any model, it can be used to compare explanations from architecturally different models on the same data. Among the hair loss studies above, the emphasis is on accuracy, and none compares explanations across a boosted-tree model and foundation models.

Table 2.1: Summary of reviewed studies

Study	Data	Models	Main result	Limitation
Khatun et al. (2022)	Survey, 610 people, Bangladesh	SVM, KNN, LR, RF, XGBoost	XGBoost 92.62%	Classical models only, no explanation
Sai et al. (2023)	Hair fall data	SVM, KNN, DT, RF, LR, ensemble	Ensemble best	No categorical-aware boosting, no explanation
Kumar et al. (2025)	2,000 records, 10 features	RF, XGBoost, CatBoost, LightGBM and others	RF 100%, CatBoost 49.5%	Suspiciously high score, CatBoost untuned
Siami and Azis (2025)	Balanced multi-factor data	LR, DT, RF, GB, XGBoost, voting	About 50% at best	Weak signal, modest accuracy
Leema et al. (2025)	Survey, 750 people	LSTM, RF, TFT, ARIMAX	TFT 97.5%	Private data, self-reported only
Shakeel et al. (2021)	Hair images	SVM, KNN	91.4% and 88.9%	Images needed, one condition
Sayyad et al. (2022)	268 images	VGG with SVM	98.31%	Small image set, one condition
Pandikumar et al. (2024)	Scalp images and lifestyle sequences	CNN, LSTM	Framework proposed	Needs images
Prokhorenkova et al. (2018)	Benchmarks	CatBoost	Beats other boosting libraries	Not health-specific
Hollmann et al. (2025)	Benchmarks	TabPFN	Strong on small tables	No hair loss data
Kong et al. (2026)	TabArena, 51 datasets	TabFM	First among default foundation models	No health data
Pandey (2026)	13 TabArena datasets	TabFM vs XGBoost, RF, TabPFN	Confirmed competitive	Four defects, memory limit

LR = Logistic Regression, DT = Decision Tree, RF = Random Forest, GB = Gradient Boosting.

2.2.1 Research Gap

Reading the studies together shows five gaps, and each one explains a choice made in this project.

1. **Only classical algorithms were compared on hair loss tables.** The survey-based studies used Logistic Regression, SVM, KNN, Random Forest, and XGBoost (Khatun et al., 2022; Sai et al., 2023; Siami & Azis, 2025). The reported accuracies range from about 50% to 100% on different datasets, so there is no reliable picture of which type of model suits hair loss data. This project therefore compares different families of models on one dataset with one fixed split.
2. **CatBoost was never properly tuned for hair loss.** CatBoost was designed for categorical features, which dominate survey data, yet it appears only once as a side comparison with a low score (Kumar et al., 2025). This project tunes CatBoost carefully so that it acts as a strong baseline.
3. **Tabular foundation models have not been tested on hair loss or on health survey data.** TabPFN and TabFM were evaluated on general benchmarks (Hollmann et al., 2025; Kong et al., 2026), and the only independent check of TabFM used TabArena datasets (Pandey, 2026). Whether such models can match a tuned boosted-tree model on a health table is not known. This is the reason TabPFN and TabFM are included and given the same data.
4. **TabFM is very new and its practical limits are unclear.** The independent reproduction found software defects and memory failures on larger tables (Pandey, 2026). Running TabFM on a real dataset and reporting its cost and limitations is a contribution in itself.
5. **Explanations are not compared across models.** The hair loss studies report accuracy and rarely explain individual predictions, and none explains a boosted-tree model and foundation models side by side. This project applies SHAP (Lundberg & Lee, 2017) to all three models so the clinical plausibility of their explanations can be compared.

Taken together, the gaps lead to the two objectives of this study: to benchmark CatBoost, TabPFN, and TabFM on the same hair fall risk dataset (Objective 1), and to explain every

model with SHAP (Objective 2). Chapter 3 describes the dataset and methods that will be used to do this.

Chapter 3: Methodology

Chapter 2 ended with five research gaps, and Chapter 1 set two objectives: to benchmark three models and to explain them. This chapter describes the data and the methods that answer them. It starts with the overall framework (Section 3.1), then the dataset and its preparation (Sections 3.2 and 3.3), the three models (3.4), the explainability method (3.5), the tools and environment (3.6 and 3.7), and the evaluation metrics (3.8). The dataset work and preprocessing have already been done, and the experiments will be carried out after the proposal is approved.

3.1 Research Framework

The study follows one pipeline from data to explanation, shown in Figure 3.1. The final dataset is cleaned and split once. The same training and test rows are then given to three models, CatBoost, TabPFN, and TabFM. To see how each model behaves when little data is available, each will also be given two smaller stratified subsets of the training rows (500 and 2,000 records) in addition to all 17,284. Their predictions are compared with the same metrics and statistical tests, and SHAP is applied to all three so that each predicted risk tier can be traced to its features.

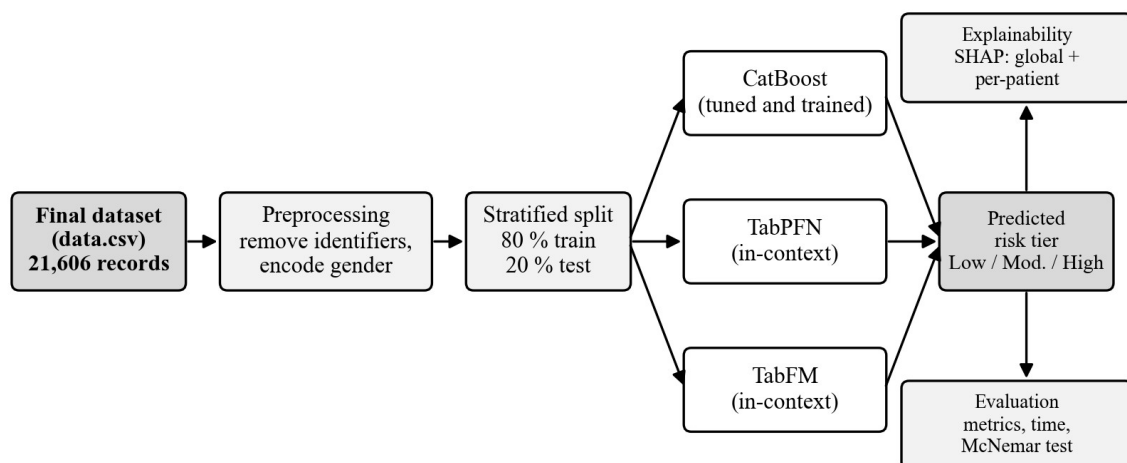


Figure 3.1: Overall research framework

The framework serves the two objectives in Chapter 1. Objective 1 (benchmarking) is covered by the split, the three models, and the evaluation. Objective 2 (transparent insight) is covered by the SHAP stage.

3.2 Dataset Description

3.2.1 Source datasets and how they are related

Three datasets were involved. The main dataset used for modelling contains all 21,606 records and all columns needed for the study. To check that this dataset is realistic and that the findings do not depend on one source only, it was compared with two public datasets. Table 3.1 lists the three datasets, and Figure 3.2 shows how they are related.

Table 3.1: The three datasets

Dataset	Source	Rows	Columns	Content	Row identifier
Dataset 1	Kaggle "Hair Loss Dataset" (Dhankour, 2023)	100,000	11	10 numeric biomarker and index columns and hair_fall (0 to 5)	None
Dataset 2	Mendeley "Dataset for evaluating hair fall causes" (Arnob et al., 2024)	716	14	Questionnaire: age, gender, 8 Yes/No health and lifestyle answers, hair fall problem, food habit	Timestamp
Final dataset (data.csv)	Prepared for this study	21,606	23	20 features and the 3-tier target hair_fall	id (1 to 21,606)

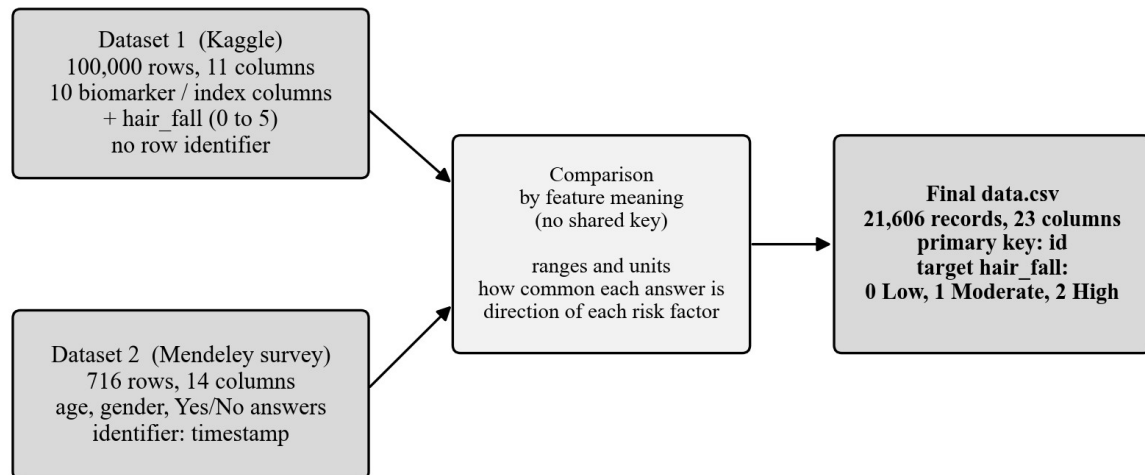


Figure 3.2: Relationship between the source datasets and the final dataset

The datasets are related by the meaning of their columns and not by a record key. Dataset 1 has no identifier at all, and Dataset 2 has only a timestamp, and the people in the three datasets are not the same individuals. A record-by-record join is therefore not possible.

Instead, each column of the final dataset was matched to the column or question that measures the same thing, as shown in Table 3.2. The primary key of the final dataset is its own id column, which runs from 1 to 21,606 and was removed before modelling.

Table 3.2: Where each column of the final dataset comes from

Final column	Dataset 1 column	Dataset 2 question	Unit or coding
age	none	What is your age?	years
gender	none	What is your gender?	Female, Male, Other
total_protein	total_protein	none	g/dL
calcium	calcium	none	mg/dL
iron	iron	none	µg/dL
vitamin_d	vitamin	none	ng/mL
alt_liver	liver_data	none	U/L
manganese	manganese	none	µg/L
body_water_content	body_water_content	none	%
stress_level	stress_level	none	0 to 40
total_keratine	total_keratine	none	0 to 100 score
hair_texture	hair_texture	none	0 to 100 score
family_hair_fall_history	none	Family member with hair fall or baldness?	0 = No, 1 = Yes
chronic_illness	none	Chronic illness in the past?	0 = No, 1 = Yes
late_night_sleep	none	Stay up late at night?	0 = No, 1 = Yes
sleep_disturbance	none	Any sleep disturbance?	0 = No, 1 = Yes
water_reason	none	Water in your area a reason for hair fall?	0 = No, 1 = Yes
chemical_use	none	Use chemicals, gel or colour in hair?	0 = No, 1 = Yes
anemia	none	Do you have anemia?	0 = No, 1 = Yes
stress	none	Do you have too much stress?	0 = No, 1 = Yes
hair_fall (target)	hair_fall (0 to 5)	Do you have a hair fall problem?	0 Low, 1 Moderate, 2 High

3.2.2 What the comparison showed

The comparison was done with a script (Datasets/comparison/dataset_comparison.py), and all numbers below come from it.

Dataset 1. Every column of Dataset 1 is spread evenly between its minimum and maximum (for example, iron runs from 0 to 499 and stress_level from 0 to 99), and the target hair_fall is spread evenly over the six values 0 to 5. Its columns also show no relationship with its own target: the Spearman correlation between every biomarker and hair_fall lies between -0.006 and $+0.004$. Dataset 1 therefore supplies the list of biomarkers and their types, but its values cannot be used directly to predict risk. In the final dataset the values lie in real clinical ranges (for example iron 10 to 217 $\mu\text{g/dL}$ and total protein 4.8 to 9.5 g/dL) and most biomarkers are related to the target (for example iron $\rho = -0.31$, stress_level $\rho = +0.33$, total_protein $\rho = -0.30$).

Dataset 2. Dataset 2 is a real survey of young people (mean age 23.9 years) and was cleaned first: one age of 218 was removed and two typing errors ("Yea" and "\No") were corrected, leaving 715 usable rows. Two checks were made against the final dataset, shown in Table 3.3. First, the risk factors are much more common in the survey (for example stress 72.0% and family history 72.7%) than in the final dataset (40.3% and 34.5%), because the survey respondents are self-selected and many were interested in hair fall. Second, and more important, the direction of each risk factor is the same in both datasets. In the survey, people with a hair fall problem more often reported every one of the eight factors than those without, and the same is true in the final dataset for all eight.

Table 3.3: Risk factors in Dataset 2 and in the final dataset (percentage points)

Risk factor	Dataset 2: how common (%)	Final: how common (%)	Dataset 2: gap, hair fall vs none	Final: gap, hair fall vs none	Same direction
Family history	72.7	34.5	+41.0	+10.1	Yes
Chronic illness	45.7	29.9	+41.1	+8.7	Yes
Late-night sleep	65.3	40.2	+44.7	+6.7	Yes
Sleep disturbance	55.4	35.2	+41.4	+6.5	Yes
Water as a reason	60.0	29.8	+51.4	+5.0	Yes
Chemical use	63.9	39.9	+40.7	+10.8	Yes
Anemia	28.7	24.6	+22.2	+8.7	Yes
Stress	72.0	40.3	+48.3	+11.2	Yes

Gap = share of people with the factor among those with hair fall minus the share among those without. For the final dataset, "hair fall" means Moderate or High risk.

The survey shows stronger gaps than the final dataset because survey respondents answered about a problem they already knew they had, while the final dataset describes graded risk. Because of these differences the final dataset is treated as a **semi-synthetic benchmark**: the associations learned by the models reflect relationships built into the data from the dermatology literature and the survey, and they are not new clinical evidence. **[CONFIRM: add one sentence on how the 21,606 records were first generated.]** This limitation is stated again when the results are discussed.

3.2.3 The final dataset

The final dataset has 21,606 records with 20 features and one target, in four groups.

1. **Demographics and heredity:** age (20 to 55 years), gender, family_hair_fall_history.
2. **Blood and body measurements:** total_protein, calcium, iron, vitamin_d, alt_liver (alanine aminotransferase, ALT), manganese, body_water_content, stress_level (0 to 40 scale), and two 0 to 100 hair condition scores, total_keratine and hair_texture.
3. **Clinical conditions (0/1):** chronic_illness, anemia, stress.
4. **Lifestyle and environment (0/1):** late_night_sleep, sleep_disturbance, water_reason, chemical_use.

The target hair_fall has three risk tiers: 0 = Low (9,723 records, 45%), 1 = Moderate (7,562, 35%), and 2 = High (4,321, 20%). Gender is Male for 12,109 records, Female for 9,485, and Other for 12.

3.3 Data Preprocessing

The following steps were applied to the final dataset before any model was run.

1. **Removal of identifiers.** The id column and the full_name column (which holds personal names in encoded form) were removed, because they do not describe health and could create false patterns. After removal, 20 features remain.

2. **Quality checks.** The dataset has no missing values and no duplicate records (ignoring id).
3. **Clinical range check.** Each blood and body measurement was compared with its normal reference range and with limits that are physiologically impossible (Table 3.4). No values were impossible, so no record was removed. Values outside the normal range were kept on purpose, because they are the information that indicates risk.
4. **Encoding.** Gender was coded 0 = Female, 1 = Male, 2 = Other. The yes/no columns were already 0/1, and the target was already coded 0, 1, 2.
5. **Stratified split.** The records were split 80% to 20% with the class shares kept equal (random seed 42): 17,284 training records and 4,322 test records, with 45% Low, 35% Moderate, and 20% High in both parts.

Table 3.4: Clinical range check of the final dataset

Measurement	Normal range	Below normal	Above normal	% outside normal	Impossible values
total_protein (g/dL)	6.0 to 8.3	406	605	4.7	0
calcium (mg/dL)	8.5 to 10.5	1,675	392	9.6	0
iron (µg/dL)	60 to 170	2,217	209	11.2	0
vitamin_d (ng/mL)	25 to 80	8,606	0	39.8	0
alt_liver (U/L)	7 to 56	2,300	403	12.5	0
manganese (µg/L)	4 to 15	1,960	208	10.0	0
body_water_content (%)	45 to 65	2,225	2,227	20.6	0

3.4 Model Selection

Three models from different families are compared: a tuned gradient boosting model (CatBoost) and two tabular foundation models that predict without training (TabPFN and TabFM). Each is shown as a block diagram with a short explanation of how it works.

3.4.1 CatBoost

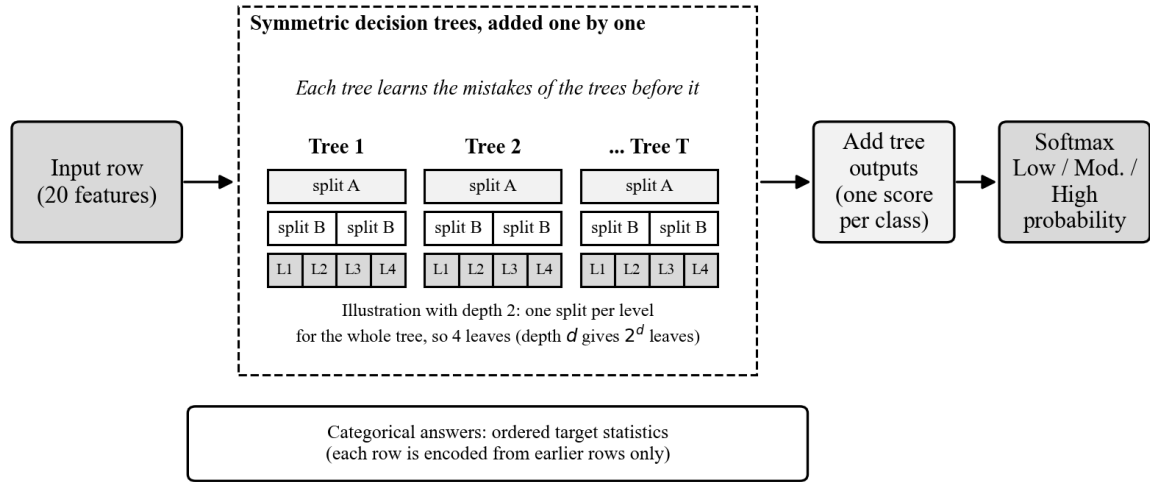


Figure 3.3: CatBoost model (gradient boosted symmetric trees)

CatBoost builds many small decision trees one after another, and each new tree h_t is trained to correct the mistakes of the ensemble F_{t-1} built so far, by minimising

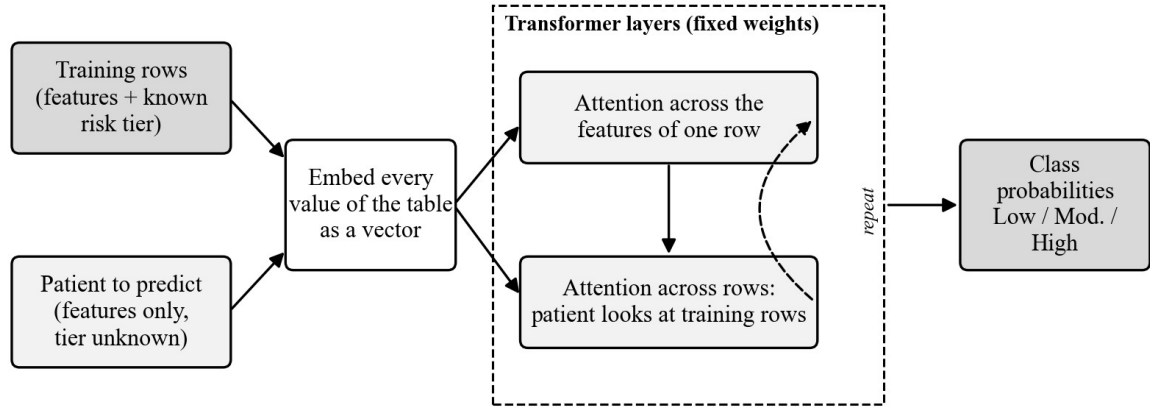
$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, F_{t-1}(x_i) + h_t(x_i)) + \Omega(h_t) \quad (\text{i})$$

where l is the multi-class cross-entropy loss, y_i is the true tier of record i , and $\Omega(h_t)$ is a penalty on the complexity of the new tree, which limits overfitting. All trees are symmetric, meaning that one split is used for a whole level of the tree, which makes the model fast and hard to overfit. The tree outputs are added to give one score s_k for each risk tier k , and a softmax turns the three scores into probabilities:

$$P(y = k | x) = \frac{e^{s_k}}{\sum_{j=1}^3 e^{s_j}} \quad (\text{ii})$$

Categorical answers are encoded using only earlier rows, which avoids target leakage (Prokhorenkova et al., 2018). CatBoost is the only model that will be trained and tuned on the hair fall data. Its tuning will be a grid search over tree depth and learning rate, scored by macro-F1 with 5-fold cross-validation on the training rows only, so that the test set stays untouched. The two foundation models will not be tuned.

3.4.2 TabPFN



One forward pass, no training on hair fall data. The weights were learned once on millions of synthetic tables.

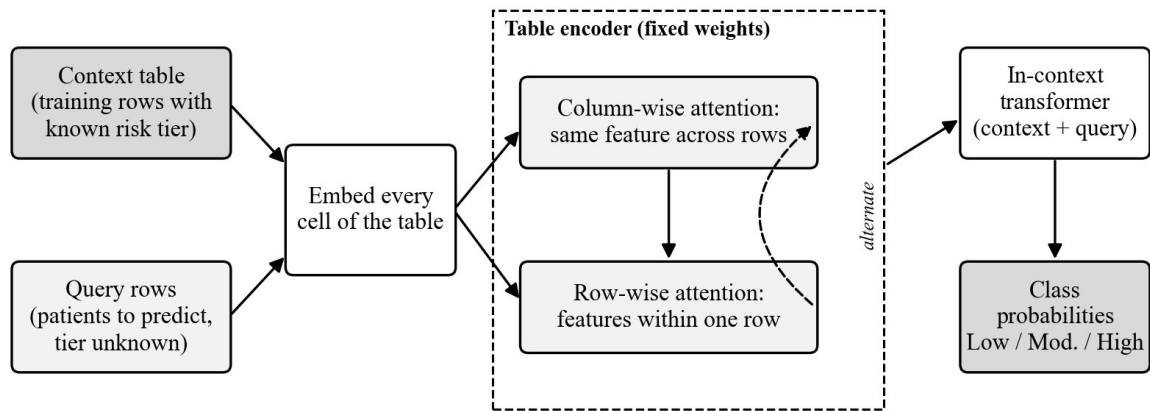
Figure 3.4: TabPFN model (in-context learning)

TabPFN is a transformer that was trained beforehand on millions of synthetic tables (Hollmann et al., 2025). To predict, it receives the training rows together with their known risk tiers and the patient to be predicted. Every value is turned into a vector, and attention layers let the model relate the features of a row and relate the patient to the training rows. The class probabilities are produced in a single forward pass:

$$P(y | x_{test}, D_{train}) = f_{\theta}(x_{test}, D_{train}) \quad (\text{iii})$$

where D_{train} is the set of training rows with their tiers and f_{θ} is the pretrained network with fixed weights θ . No weights are changed, so nothing is trained on the hair fall data. TabFM works in the same way, with its own network in place of f_{θ} .

3.4.3 TabFM



About 400 million parameters, pretrained once on synthetic tables. No training or tuning on hair fall data.

Figure 3.5: TabFM model (zero-shot tabular foundation model)

TabFM is a model of about 400 million parameters developed by Google Research and pretrained on synthetic tables (Kong et al., 2026). Like TabPFN, it takes the training rows as context and predicts in one forward pass without tuning. Its table encoder alternates attention along columns and along rows, and an in-context transformer then combines the context rows with the query rows to give the class probabilities. It was published in 2026 and has been tested by others only on general benchmarks (Pandey, 2026), so how it behaves on real health data is one of the open questions of this study.

3.5 Explainability with SHAP

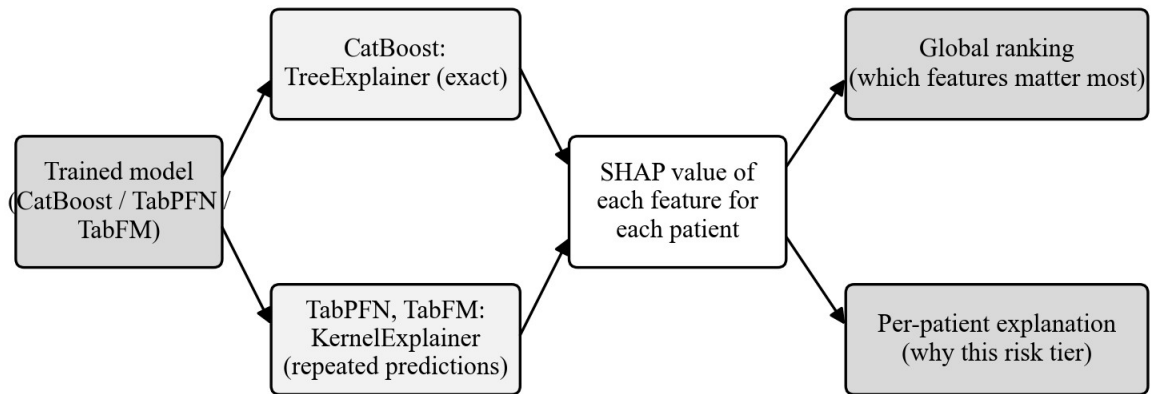


Figure 3.6: SHAP explainability workflow

SHAP splits a prediction into one contribution per feature, based on Shapley values from game theory (Lundberg & Lee, 2017). The contributions add up to the difference between the model's output and its average output:

$$f(x) = \phi_0 + \sum_{j=1}^{20} \phi_j \quad (\text{iv})$$

where $f(x)$ is the model output for a patient x , ϕ_0 is the average prediction and ϕ_j is the contribution of feature j (for example iron or stress_level) to the predicted risk tier.

- **CatBoost:** TreeExplainer, which uses the tree structure to compute exact values, will be applied to a sample of 1,000 test records.
- **TabPFN and TabFM:** KernelExplainer, which estimates the values by repeatedly changing the inputs and predicting again. Because every repeat is a full forward pass of a large model, it will be applied to a small sample of test records (a few tens) with a small k-means background set.

The results will be a global ranking of the features (which matter most overall) and per-patient explanations (why this person received this tier). The rankings of the three models will be compared with each other and with the known biology from Chapter 2.

3.6 Tools and Technologies

Table 3.5 lists the tools used for the work, with the versions recorded in the local environment.

Table 3.5: Software tools

Purpose	Tool	Version
Language	Python	3.12.14
Data handling	pandas, NumPy	3.0.6, 2.5.3
Split, metrics, cross-validation	scikit-learn	1.9.1
Statistical tests	statsmodels (McNemar's test, Wilson intervals)	0.15.0
Gradient boosting model	CatBoost	1.2.10
Foundation model 1	TabPFN	9.0.0
Foundation model 2	TabFM (weights from Hugging Face: google/tabfm-1.0.0-pytorch)	1.0.1
Explainability	SHAP	0.52.0
Deep learning backend	PyTorch	2.14.0
Charts	Matplotlib	3.11.2
Workflow	Local lab application (notebook-style steps) with PostgreSQL and MinIO	—
GPU runs	Kaggle notebooks	—

Package versions used on Kaggle are recorded at the start of each notebook and reported with the results.

3.7 Experimental Environment

The work is divided between two environments (Figure 3.7).

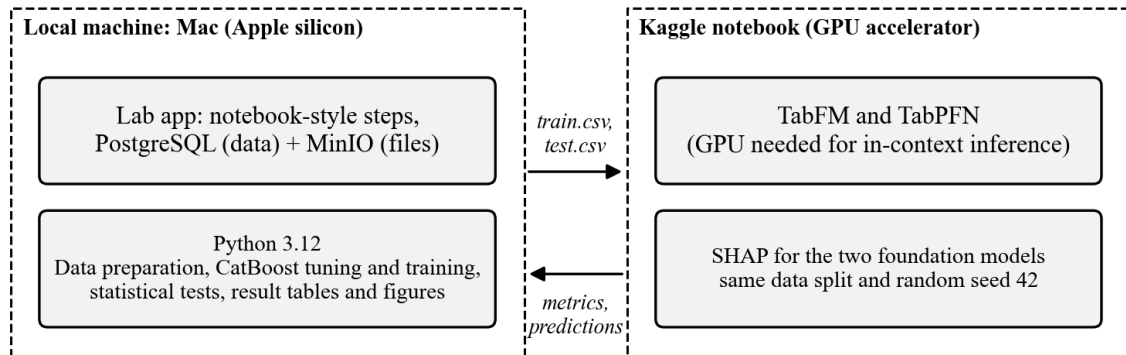


Figure 3.7: Experimental environment

Local machine. Data preparation and the split have already been done on an Apple-silicon Mac (macOS, 17.2 GB memory, Apple GPU) with Python 3.12, and CatBoost tuning and training, the statistical tests, and the result tables and figures will be run there as well. The steps are run one by one in a local lab application that stores the data in PostgreSQL and the outputs in MinIO, so every run is saved and can be reopened.

Kaggle notebooks. TabFM and TabPFN need a GPU to make their predictions in reasonable time, and TabFM is the largest model (about 6.5 GB of weights). These runs will therefore be done in Kaggle notebooks, which offer more GPU capacity than Google Colab. The train and test files created on the local machine are uploaded to the notebook, and the metrics and predictions are downloaded back for comparison. SHAP for the two foundation models will also be run there.

To keep the comparison fair, all three models use the same split and the same random seed (42), and the environment of every session (Python, GPU, memory, package versions) is recorded and reported.

3.8 Performance Evaluation Metrics

Model performance will be measured on the same held-out test set for every model. Because the three risk tiers are not equally common (45%, 35%, 20%), the class-averaged (macro) versions of the metrics are used, so that the smaller High tier counts as much as the Low tier. Here TP_k , FP_k , and FN_k are the true positives, false positives, and false negatives of tier k . Each equation is numbered on the right and cited in the text as Equation (i), (ii), and so on.

Accuracy is the share of test records whose tier is predicted correctly, where N is the number of test records:

$$\text{Accuracy} = \frac{TP_1 + TP_2 + TP_3}{N} \quad (\text{v})$$

Precision shows how trustworthy a predicted tier is. It is computed for each tier k and then averaged over the three tiers (macro-precision):

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \quad \text{Macro Precision} = \frac{1}{3} \sum_{k=1}^3 \text{Precision}_k \quad (\text{vi})$$

Recall shows how many patients of a tier are found (macro-recall is the average over the three tiers):

$$\text{Recall}_k = \frac{TP_k}{TP_k + FN_k}, \quad \text{Macro Recall} = \frac{1}{3} \sum_{k=1}^3 \text{Recall}_k \quad (\text{vii})$$

Macro-F1 is the harmonic mean of precision and recall for each tier, averaged over the tiers:

$$F1_k = \frac{2 \text{Precision}_k \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}, \quad \text{Macro F1} = \frac{1}{3} \sum_{k=1}^3 F1_k \quad (\text{viii})$$

ROC-AUC is the area under the receiver operating characteristic (ROC) curve of each tier against the other two (one-versus-rest), averaged over the three tiers. It measures how well the predicted probabilities rank patients, independent of any cut-off:

$$\text{Macro ROC AUC} = \frac{1}{3} \sum_{k=1}^3 AUC_k \quad (\text{ix})$$

- **Confusion matrix:** shows which tiers are mixed up. This matters here because the tiers are ordered, and confusing Low with High is a worse mistake than confusing neighbouring tiers.

Statistical significance. McNemar's test will be applied to every pair of models on the test set, to decide whether a difference in the number of correct predictions is larger than chance. Let b be the number of records that only the first model predicts correctly and c the number that only the second model predicts correctly:

$$\chi^2 = \frac{(lb - cl - 1)^2}{b + c} \quad (\text{x})$$

with one degree of freedom (the -1 is the continuity correction). The accuracy of each model is also given with a 95% Wilson confidence interval, where $\hat{p}$ is the accuracy, n is the number of test records, and $z = 1.96$:

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \quad (\text{xi})$$

The next chapter states what this study expects these methods to show and sets out the schedule for carrying them out.

Chapter 4: Expected Outcome and Working Schedule

Chapter 3 described the methods that will be used. This chapter states what those methods are expected to show (Section 4.1) and when each part of the work will be done (Section 4.2).

4.1 Expected Outcomes

The expected outcomes follow directly from the research gaps found in the literature review (Section 2.2.1) and from the two objectives in Section 1.3. Each gap is matched below with what this study expects to deliver. The expectations are stated as aims to be tested, not as findings, because the experiments are still to be carried out.

1. A fair benchmark of different model families on hair fall data (gap 1, Objective 1).

Earlier hair fall studies compared only classical algorithms, on different datasets, with accuracies ranging from about 50% to 100% (Khatun et al., 2022; Kumar et al., 2025; Siami & Azis, 2025). This study is expected to give a comparison that can be trusted, because CatBoost, TabPFN, and TabFM use the same 21,606 records, the same train and test split, the same random seed, and the same metrics.

2. A properly tuned gradient boosting baseline (gap 2, Objective 1). CatBoost scored only 49.5% in the one earlier hair fall study that included it, apparently without tuning (Kumar et al., 2025). Here CatBoost is tuned with a grid search and cross-validation on the training data only, so it is expected to be a strong and fair baseline. Any advantage of a foundation model over it would then be meaningful.

3. The first test of tabular foundation models on hair fall risk (gap 3, Objective 1).

TabPFN and TabFM performed well on general benchmarks (Hollmann et al., 2025; Kong et al., 2026), and this study expects them to be competitive with tuned CatBoost on hair fall data without any tuning, especially when only a small part of the training data is used. This is an expectation taken from the benchmark literature, and the study is designed so that it can also show the opposite. A difference between the models is accepted only if McNemar's test shows that it is larger than chance.

4. A record of the practical limits of TabFM (gap 4, Objective 1).

An independent reproduction reported software defects and memory failures on larger tables (Pandey, 2026). The study is expected to document how TabFM behaves on a real health dataset, including

how much training data it can take, how long it runs, and what problems appear, so that later researchers know what to expect.

5. A side-by-side comparison of explanations (gap 5, Objective 2). SHAP is expected to show which measurements and habits drive each model's predicted tier, and to show whether the three models agree. The factors that the models rely on are expected to match what Chapter 2 describes as the biological causes of hair loss, such as iron level, stress, and family history. Agreement would support the use of the models as decision support. Disagreement would be reported as a finding and not hidden.

Table 4.1 summarises how each expected outcome will be checked.

Table 4.1: Expected outcomes and how they will be checked

Research gap (Section 2.2.1)	Expected outcome	How it will be checked	Method in Chapter 3
1. Only classical models, different datasets	Fair three-model benchmark	Same data, split, seed, and metrics	Sections 3.3, 3.8
2. CatBoost not tuned	Strong tuned baseline	Grid search with cross-validation on training data	Section 3.4.1
3. Foundation models not tested on hair fall	Competitive results without tuning, to be confirmed	Macro-F1 and ROC-AUC, McNemar's test, Wilson intervals	Sections 3.4.2, 3.4.3, 3.8
4. TabFM limits unknown	Documented limits of TabFM	Recorded settings, problems, and run conditions	Section 3.7
5. Explanations not compared	Comparable SHAP explanations for all three models	Global ranking and per-patient explanations	Section 3.5

Beyond the individual outcomes, the study is expected to show whether explainable, data-efficient tabular modelling can be a practical alternative to heavily tuned gradient boosting for structured health risk prediction, and to leave a documented and reproducible pipeline that others can rerun on their own data.

4.2 Working Schedule

The work is planned over six months, with the schedule shown as a Gantt chart in Figure 4.1. The tasks on the left are split into what has already been done up to this proposal (black bars)

and what will be done after the proposal is approved (hatched bars). The completed tasks are the problem formulation, the literature review, the comparison and preparation of the datasets, the preprocessing and split, and the setup of the local environment and pipeline code. The planned tasks follow the order of the methodology in Chapter 3: model training, the training-size experiment, SHAP, evaluation, discussion, and thesis writing.

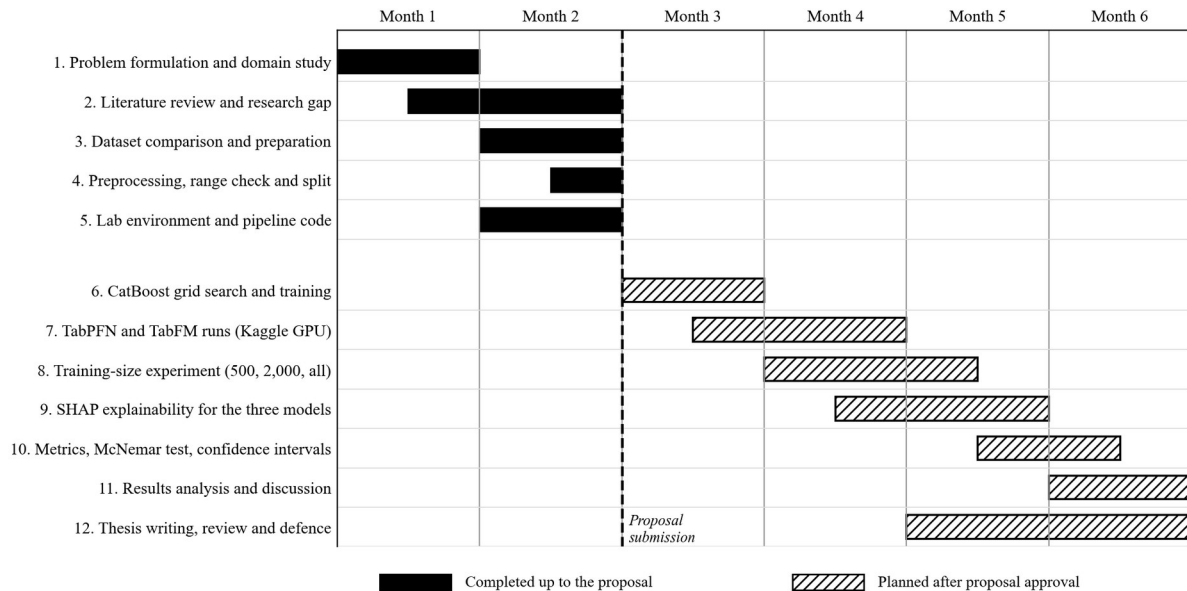


Figure 4.1: Working schedule (Gantt chart)

After the proposal is approved, the planned tasks will be carried out in this order, and the results will be reported and discussed against the expected outcomes of Section 4.1 in the final dissertation.

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